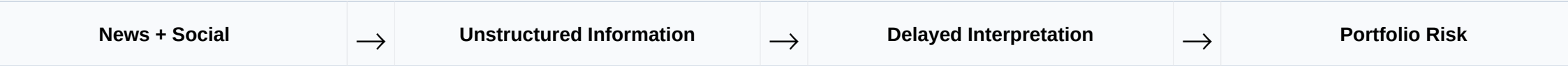


Financial Markets Are Information-Rich but Signal-Poor

Markets continuously generate large volumes of financial information, but the difficult step is turning that unstructured information into **timely, explainable portfolio decisions**.



Hackathon objective: build a unified AI/NLP risk engine that transforms text into sentiment, event and impact signals — then demonstrates a downstream tactical portfolio decision.

FinRisk AI addresses the gap between information overload and actionable risk intelligence.

FinRisk AI

Real-Time Financial Risk Intelligence & Tactical Index Rebalancing



Ingest Public financial information	Understand Financial-domain sentiment	Classify Event + impact severity
Aggregate Combine signals into risk intelligence	Decide INCREASE / HOLD / REDUCE	Explain Expose the reasoning behind each action

From Raw Text to Portfolio Decision



Backend Python + FastAPI Pandas / NumPy	NLP FinBERT / Transformers Financial sentiment	Decision Layer Explainable scoring Weight constraints
Frontend React + Vite Recharts visualization	Output Structured JSON /signals + /rebalance	Reliability Public/synthetic fallback for deterministic demos

The architecture is API-first: the same structured signals power the dashboard and can be consumed by downstream systems.

Unstructured Text → Machine-Readable Risk

Illustrative risk record

NVDA	-0.78	Geopolitical	8 / 10	CRITICAL
Ticker	Sentiment	Event	Impact	Risk

- **1. Sentiment** — FinBERT maps financial language to a score from approximately -1 to +1.
- **2. Event** — transparent event rules classify categories such as geopolitical, macroeconomic, credit, M&A; and product events.
- **3. Impact** — severity combines sentiment strength, event category and high-risk language on a 1–10 scale.
- **4. Risk** — the structured record becomes an explainable signal for downstream decisions.

Tactical Index Rebalancing

A mock 15-stock index starts from near-equal weights. Risk signals then drive explainable portfolio actions while weight constraints limit concentration.

Stock	Base Weight	Sentiment	Impact	New Weight	Action
MSFT	6.67%	+0.187	3.8	7.77%	INCREASE
AAPL	6.67%	0.000	3.4	6.84%	HOLD
Example negative signal	6.67%	Negative	High	Lower	REDUCE

Signal	→	Risk-Adjusted Score	→	Weight Update	→	INCREASE / HOLD / REDUCE
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Design principle: the portfolio layer is not a black box — every weight change can be traced back to sentiment, impact and signal aggregation.

Measured Prototype Validation

75	15	-1 to +1	1–10
news signals processed in a live run	mock index stocks	sentiment scale	impact scale

- ✓ Live ingestion endpoint returned HTTP 200.
- ✓ Structured /signals endpoint returned processed risk records.
- ✓ Tactical /rebalance endpoint returned weight and action outputs.
- ✓ FinBERT model loaded successfully after first-time model download.

Important: no unsupported accuracy or performance claims are presented. The current validation focuses on end-to-end functionality and measured outputs.

From Information Overload to Actionable Risk Intelligence

01 Multi-source architecture	02 Financial-domain sentiment	03 Explainable event + impact scoring
04 Source confidence + recency concepts	05 Automatic tactical rebalancing	06 API-first decision support

Domain impact: FinRisk AI turns scattered financial text into structured risk intelligence and connects that intelligence directly to a transparent portfolio action layer.

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